SUPPLEMENTARY MATERIAL LEGEND

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Supplemental Appendix S1

Search strategy for Embase

1: ‘percutaneous coronary intervention’:ti,ab

2: ‘primary PCI’:ti,ab

3: ‘pci’:ti,ab OR ‘ptca’:ti,ab

4: 1 OR 2 OR 3

5: ‘myocardial infarct*’:ti,ab OR ‘stemi’:ti,ab OR ‘ami’:ti,ab

6: ((corona* OR corono*) NEXT/1 (virus* or viral* or virinae*)):ti,ab

7: (coronavirus* OR coronavirus* OR coronavirinae* OR ‘2019-ncov’ OR ‘2019 cov’ OR ncov2019 OR ncov-2019 OR covid-19 OR covid19 OR corvid-19 OR corvid19 OR wncov OR wn-cov OR hcov-19 OR hcov19 OR ‘2019 novel’ OR ‘ncov’ OR n-cov OR ‘sars-cov-2’ OR ‘sarscov-2’ OR sarscov2 OR sars-cov2 OR sarscov19 OR sars-cov19 OR sars-cov-19 OR ‘sarsr cov’ OR ncovor OR ncorona* OR ncorono* OR ncovwuhan* OR ncovhubei OR ncovchina* OR ncovchinese*):ti,ab

8: 6 OR 7

9: 4 AND 5 AND 8

Search strategy for Medline

1: (percutaneous coronary intervention OR primary PCI OR PCI or angioplast* OR PTCA).tw.

2: (myocardial infarct* OR STEMI or AMI).tw.

3:(corona-virus*.tw OR corono-virus*.tw OR coronavirus*.tw OR coronovirus*.tw OR coronavirinae*.tw OR Wuhan*.tw OR Hubei*.tw OR Huanan.tw OR "2019 nCoV".tw OR 2019nCoV.tw OR 2019 CoV.tw OR nCoV2019.tw OR "nCoV 2019".tw OR "COVID 19".tw OR COVID19.tw OR "CORVID 19".tw OR CORVID19.tw OR "WN CoV".tw OR

WNCoV.tw OR "HCoV 19".tw OR HCoV19.tw OR CoV.tw OR "2019 novel*".tw OR Ncov.tw OR "n cov".tw OR "SARS CoV 2".tw OR "SARSCoV 2".tw OR "SARS-CoV-2".tw OR "SARSCoV-2".tw OR "SARSCoV2".tw OR "SARS CoV2".tw OR SARSCov19.tw OR "SARS Cov19".tw OR "SARSCov 19".tw OR "SARS Cov 19".tw OR Ncovor.tw OR Ncorona*.tw OR Ncorono*.tw OR NcovWuhan*.tw OR NcovHubei*.tw OR NcovChina*.tw OR NcovChinese*.tw OR SARSr-cov).tw

4: 1 AND 2 AND 3

Supplemental Table S1 Door-to-Balloon Time Definition

Author	Door-to-Balloon Time Definition
Boeddinghaus et al	Not Defined
Cammalleri et al	Time interval between admission to ED[a] and successful wire crossing
Çinier et al	Time interval between arrival at hospital and balloon inflation
Claeys et al	Time interval between admission to PCI[b] center and balloon inflation
Clifford et al	Not defined
Chew et al	Time interval between arrival at hospital and first device deployment[c]
De Luca et al	Not defined
Erol et al	Time interval between arrival at study center and balloon inflation
Gramegna et al	Time interval between admission to hospital and wire crossing culprit lesion site
Haddad et al	Time interval between arrival at hospital and first device activation
Kobo et al	Not defined
Kwok et al	Not defined
Little et al	Time interval between arrival at PCI[b] center and first coronary intervention restoring perfusion
Nan et al	Not defined
Natarajan et al	Not defined
Popovic et al	Not defined
Reinstadler et al	Not defined
Song et al	Not defined
Soylu et al	Time interval between arrival at PCI center and successful wire crossing

[a]**ED:** Emergency Department

[b]**PCI:** Percutaneous Coronary Intervention

[c]Authors defined first device deployment as either balloon inflation, manual thrombectomy or direct stenting

Supplemental Table S2 Summary of Included Studies

Author	Country	Cohort	Time Period	Caseload Reduction	N	Age	Male (%)	HTN (%)	T2DM (%)	Dyslipidaemia (%)	Smoking (%)	Past MI (%)	Past PCI (%)	Past CABG (%)	Anterior (%)	Killip≥3 (%)	Out of Hosp CA (%)	Cardiogenic Shock (%)	QUIPS Rating	NOS Score
Ayad et al[50]	Egypt	CP	1 Feb – 30 Oct 2020	25.8%	270	57.1	81.5	39.6	35.2	45.2	45.6	-	-	-	47.4	-	-	-	Moderate	8
		PC	1 Feb – 30 Oct 2019		364	58.9	85.7	42.9	35.7	50.0	57.1	-	-	-	59.3	-	-	-		
Boeddinghaus et al[57]	Switzerland	CP	1 Mar – 30 Apr 2020	NC	64	-	-	-	-	-	-	-	-	-	-	-	-	-	Low	7
		PC	1 Jan – 29 Feb 2020		63	-	-	-	-	-	-	-	-	-	-	-	-	-		
Cammalleri et al[35]	Italy	CP	1 – 31 Mar 2020	63.0%	13	65.0	84.6	53.8	23.1	61.5	76.9	5.7	8.6	0.0	46.2	-	-	-	Moderate	6
		PC	1 – 31 Mar 2019		35	62.0	88.6	65.7	22.9	31.4	54.3	7.7	0.0	0.0	34.3	-	-	-		
Çinier et al[45]	Turkey	CP	5 Mar – 6 Apr 2020	48.2%	90	59.3	81.1	45.6	26.7	14.4	47.8	-	-	-	-	-	-	-	Low	8
		PC	5 Mar – 6 Apr 2019		174	63.7	85.6	50.6	34.5	14.9	67.2	-	-	-	-	-	-	-		
Claeys et al[49]	Belgium	CP	13 Mar – 3 Apr 2020	2017: 150, 2018: 152, 2019: 177, 2020: 116[a]	116	63.0	80.0	48	27	-	-	-	-	-	47	-	-	-	Moderate	7
		PC	13 Mar – 3 Apr 2017/18/19		479	63.0	74.0	47	15	-	-	-	-	-	45	-	-	-		
Clifford et al[38]	Canada	CP	15 Nov 2019 – 16 Mar 2020	19.0%	193	65	70.0	51.0	28.0	45.0	40.0	12.0	13.0	4.0	-	-	-	-	Low	7
		PC	17 Mar – 16 Jul 2020		238	64	71.0	52.0	23.0	42.0	48.0	9.0	9.0	3.0	-	-	-	-		
Chew et al[23]	Singapore	CP	7 Feb – 31 Mar 2020	NC	95	59.0	56.8	53.7	35.8	58.9	55.8	8.4	12.6	0.0	-	12.6	9.5	-	Low	8
		PC	1 Oct 19 – 6 Feb 2020		208	57.0	64.4	54.8	36.5	69.7	58.7	10.6	13.0	0.0	-	13.1	1.9	-		
De Luca et al[58]	Europe	CP	1 Mar – 30 Apr 2020	18.9%	2811	64.0	74.5	53.4	18.4	42.5	41.1	9.7	12.6	2	45.5	-	7.0	8.8	Low	9
		PC	1 Mar – 30 Apr 2019		3484	64.0	73.7	54.7	18.1	41.5	41.3	9.4	12.6	1.7	45.7	-	6.6	7.7		
Erol et al[47]	Turkey	CP	17 Apr – 2 May 2020	31.2%	485	58	79.8	45.6	31.1	22.3	51.1	-	-	-	49.1	-	-	-	Moderate	8
		PC	1 – 15 Nov 2018		711	60	77.6	38.4	28.3	67.0	56.4	-	-	-	48.1	-	-	-		
Fileti et al[36]	Italy	CP	10 Mar – 10 Apr 2020	23.4%	33	-	-	-	-	-	-	-	-	-	-	-	-	-	Low	6
		PC	10 Mar – 10 Apr 2019		31	-	-	-	-	-	-	-	-	-	-	-	-	-		
Gramegna et al[20]	Italy	CP	21 Feb – 1 Apr 2020	NC	26	66.0	73.1	53.8	26.9	57.7	30.8	-	26.9	7.7	-	-	-	19.2	Moderate	6
		PC	21 Feb – 1 Apr 2018/19		21	68.0	85.7	52.4	19.0	38.1	19.0	-	4.8	4.8	-	-	-	9.5		
Haddad et al[39]	Canada	CP	15 Mar – 15 May 2020	11.67%	53	60.8	83.0	41.5	11.3	37.7	35.8	-	5.7	1.9	50.9	-	-	-	Low	7
		PC	15 Mar – 15 May 2019		60	69.5	70.0	50.0	21.7	46.7	41.7	-	18.3	1.7	46.7	-	-	-		
Hannan et al[40]	USA	CP	15 Mar – 30 Apr 2020	24% per week	463	61.9	74.1	-	27.9	-	-	-	16.6	-	-	-	-	6.5	Moderate	9
		PC	1 Jan 2019 – 14 Mar 2020		6584	63	73.8	-	29.9	-	-	-	17.0	-	-	-	-	8.3		
Hauguel-Moreau et al[42]	France	CP	17 Feb – 26 Apr 2020	65.0%	16	-	-	-	-	-	-	-	-	-	-	-	-	-	Moderate	6
		PC	17 Feb – 26 Apr 2018/19		63	-	-	-	-	-	-	-	-	-	-	-	-	-		
Huang et al[34]	China	CP	1 Feb – 15 Apr 2020	NC	31	61	80.6	51.6	25.8	22.6	58.1	-	-	-	54.8	6.5	-	-	Moderate	6
		PC	1 Jan – 31 Dec 2019		31	60	80.6	58.1	19.4	25.8	64.5	-	-	-	48.4	3.2	-	-		
Kitahara et al[53]	Japan	CP	7 Apr – 14 Aug 2020	NC	63	70	66.7	57.1	25.4	73.0	36.5	12.7	9.5	1.6	39.7	25.4	-	-	Moderate	8
		PC	1 Jan – 6 Apr 2020		359	72	72.1	70.5	37.9	70.5	34.0	8.1	9.2	2.0	50.1	17.0	-	-		
Kobo et al[52]	Israel	CP	20 Mar – 20 Apr 2020	22.0%	107	63.0	84.1	52.3	34.6	58.9	63.6	-	-	-	-	-	-	-	Low	9
		PC	20 Mar – 20 Apr 2019		136	61.0	81.6	47.8	29.4	53.7	61.0	-	-	-	-	-	-	-		
Kwok et al[21]	UK	CP	1 Jan – 30 Apr 2020	43.0%	683	64.0	72.9	39.4	16.4	32.8	38.1	11.4	11.8	2.3	-	-	-	-	Low	9
		PC	1 Jan – 30 Apr 2017/18/19		33255	63.0	74.7	43.2	18.5	34.9	40.9	12.4	12.5	2.4	-	-	-	-		
Little et al[22]	UK	CP	1 Mar – 30 Apr 2020	21.0%	348	63	80.0	51.2	24.7	35.6	41.6	12.4	12.4	1.2	-	-	10.8	13.5	Moderate	8

		PC	1 Mar – 30 Apr 2019		440	63	78.0	49.3	24.1	35.0	55.7	13.6	14.3	2.7	-	-	8.9	12.5		
Mengal et al[54]	Pakistan	CP	1 Mar – 30 Apr 2020	25.9%	1139	55.52	80.7	41.4	34.7	-	24.7	-	-	-	-	-	-	-	Low	6
		PC	1 Mar – 30 Apr 2019		1537	52.72	78.4	38.8	35.1	-	26.2	-	-	-	-	-	-	-		
Nan et al[24]	China	CP	23 Jan – 31 Mar 2020	NC	60	71.0	61.7	40	45	26.7	30.0	13.3	10	1.7	68.3	60	-	-		
		PC	1 Aug 19 – 22 Jan 2019		183	66.7	61.7	63.4	45.4	37.2	38.8	12.6	14.2	0.5	54.6	40.4	-	-	Moderate	6
Natarajan et al[41]	Canada	CP	1 Jan – 10 May 2020	11.6%	1352	-	-	-	-	-	-	-	-	-	-	-	-	-	Moderate	6
		PC	1 Jan – 10 May 2019		1555	-	-	-	-	-	-	-	-	-	-	-	-	-		
Popovic et al[43]	France	CP	26 Feb – 10 May 2020	NC	83	62.6	56.3	43.4	19.3	37.8	53.0	-	-	-	-	-	16.9	9.6		
		PC	2008 – 2017		1552	59.6	76.1	40.1	14.9	37.1	58.0	-	-	-	-	-	11.6	7.7	Moderate	6
Rangé et al[44]	France	CP	15 Mar – 14 Apr 2020	12.2% per month	122	62.93	70.5	45.8	19.0	28.2	39.7	4.1	9.0	0.0	46.7	-	-	5.7	Low	7
		PC	15 Jan – 14 Mar 2020		1942	63.6	76.1	42.2	14.9	33.8	38.3	7.0	12.1	1.8	42.3	-	-	2.9		
Reinstadler et al[48]	Austria	CP	16 Mar – 5 Apr 2020	NC	63	64	79.0	60.0	22.0	84.0	46.0	1.0	-	3.0	50.1	-	-	-	Moderate	7
		PC	24 Feb – 15 Mar 2020		100	61	69.0	65.0	18.0	79.0	42.0	1.5	-	3.0	46.0	-	-	-		
Rodríguez-Leor et al[56]	Spain	CP	16 Mar – 14 Apr 2020	22.7%	1009	63.1	78.4	51.9	22.6	46.7	44.6	-	-	-	-	9.2	-	4.1		
		PC	1 Apr – 30 Apr 2019		1305	63.7	78.4	50	25.2	45.8	45.7	-	-	-	-	9.9	-	4.1	Low	9
Salarifar et al[51]	Iran	CP	29 Feb – 30 Apr 2020	21.9%	178	58.8	77.0	78.1	45.5	53.4	41.6	-	-	-	-	-	-	-	Moderate	6
		PC	1 Mar – 30 Apr 2020		146	59.82	78.1	52.2	44.5	54.8	46.6	-	-	-	-	-	-	-		
Siudak et al[55]	Poland	CP	13 Mar – 13 May 2020	NC	97	-	-	-	-	-	-	-	-	-	-	-	-	-		
		PC	13 Mar – 13 May 2019		970	-	-	-	-	-	-	-	-	-	-	-	-	-	Low	9
Song et al[32]	China	CP	24 Jan – 31 Mar 2020	23.2%	73	61.6	80.8	58.9	21.9	53.4	49.3	9.6	9.6	2.7	43.8	8.2	-	-		
		PC	24 Jan – 31 Mar 2019		95	60.6	71.6	59	23.2	53.7	57.9	5.3	9.5	2.1	44.2	3.1	-	-	Low	8
Soylu et al[46]	Turkey	CP	10 Mar 2020 onwards	NC	82	61.2	74.4	70.7	34.1	-	37.8	12.0	-	-	39.0	-	-	-	Moderate	6
		PC	Before 13 Jan 2020		83	60.7	73.5	66.0	32.5	-	44.5	8.4	-	-	42.2	-	-	-		
Tomasoni et al[37]	Italy	CP	21 Feb – 10 Apr 2020	33.0%	34	66.0	73.5	64.7	23.5	20.6	44.1	17.6	20.6	0.0	58.8	14.7	-	-		
		PC	3 Jan – 10 Feb 2020		51	65.0	74.5	49	19.6	41.2	33.3	11.8	11.8	2.0	43.1	15.7	-	-	Moderate	7
Xiang et al[4]	China	CP	24 Jan – 20 Feb 2020	26.8%	10296	62.4	76.0	-	-	-	-	-	-	-	-	-	-	-		
		PC	27 Dec 2019 – 23 Jan 2020		14070	62.9	75.3	-	-	-	-	-	-	-	-	-	-	-	Low	8

CP = COVID period; **PC** = Pre-Covid period; **T2DM** = Type 2 Diabetes Mellitus; **MI** = Myocardial Infarction; **PCI** = Percutaneous Coronary Intervention; **CABG** = Coronary Artery Bypass Graft; **CA** = Cardiac Arrest; **QUIPS** = Quality in Prognosis Studies; **HTN =** Hypertension; **NC =** Not Comparable; **ª =** Study reported a monthly caseload comparison of the years 2017 to 2019 versus 2020

Supplemental Table S3. Summary of results, comparing outcomes during the COVID-19 pandemic to the PreCOVID-19 period

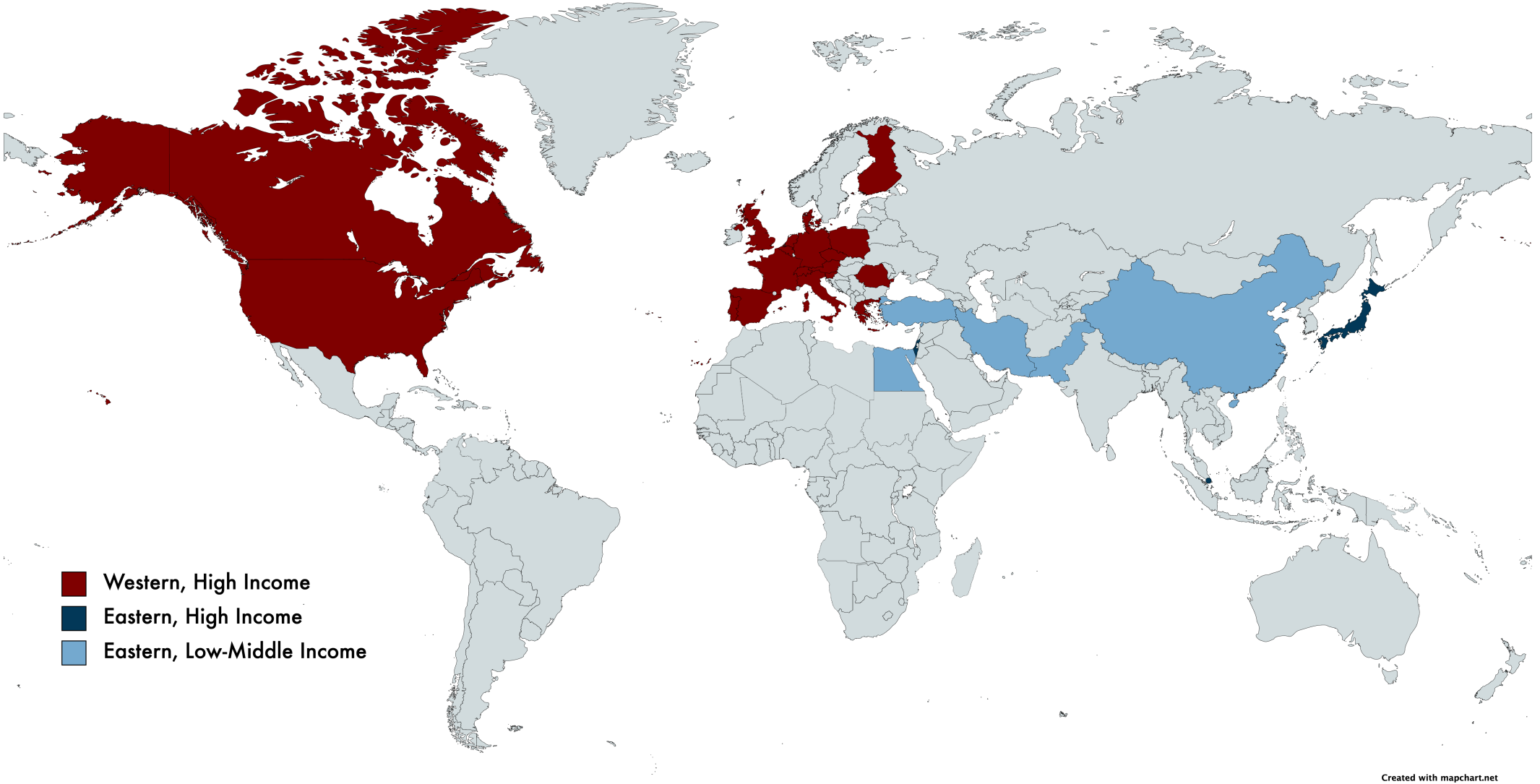
Binary Outcomes

	Overall Studies		Western Centres		Eastern Centres		High Income Countries		Non-High-Income Countries	
	OR (95% CI)	***p*-Value**	**OR (95% CI)**	***p*-Value**	**OR (95% CI)**	***p*-Value**	**OR (95% CI)**	***p*-Value**	**OR (95% CI)**	***p*-Value**
Primary Outcomes										
In Hospital Mortality	1.27 (1.09 to 1.49)	**0.002**	1.19 (0.95 to 1.49)	0.13	1.41 (1.09 to 1.83)	**0.01**	1.17 (0.95 to 1.44)	0.13	1.52 (1.13 to 2.05)	**0.006**
Secondary Outcomes										
Cardiogenic Shock	1.03 (0.86 to 1.23)	0.75	-	-	-	-	-	-	-	-
Major Adverse Cardiac Events	1.54 (1.18 to 2.00)	**0.001**	-	-	-	-	-	-	-	-
TIMI Grade <3	1.60 (1.17 to 2.21)	**0.004**	-	-	-	-	-	-	-	-

Continuous Outcomes

	Overall Studies		Western Centres		Eastern Centres		High Income Countries		Non-High-Income Countries	
	WMD (95% CI)	***p*-Value**	**WMD (95% CI)**	***p*-Value**	**WMD (95% CI)**	***p*-Value**	**WMD (95% CI)**	***p*-Value**	**WMD (95% CI)**	***p*-Value**
Primary Outcome										
Door-to-Balloon Time (mins)	8.10 (3.90 to 12.30)	**0.0002**	4.75 (0.68 to 8.83)	**0.02**	14.55 (2.88 to 26.22)	**0.01**	4.52 (0.80 to 8.25)	**0.02**	19.64 (4.61 to 34.68)	**0.03**
Secondary Outcomes										
Onset-to-Door Time (mins)	38.22 (12.67 to 63.77)	**0.003**	-	-	-	-	-	-	-	-
LVEF on Discharge (%)	- 4.20 (-1.19 to -7.21)	**0.006**	-	-	-	-	-	-	-	-
Mean ICU Stay (days)	0.70 (0.40 to 0.99)	**<0.0001**	-	-	-	-	-	-	-	-
Mean Hospital Stay (days)	0.24 (-0.46 to 0.93)	0.51	-	-	-	-	-	-	-	-

ICU: Intensive Care Unit; **LVEF**: Left Ventricular Ejection Fraction; **TIMI**: Thrombolysis in Myocardial Infarction grading system



Supplemental Figure S1. Classification of the included studies from low-middle or high income countries based on World Bank classification of countries by income.